

LAUPHEIM, Germany

WMO: 108370

Lat: 48.2175N

Lon: 9.9097E

Elev: 1804

StdP: 13.76

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	16.0	8.4	9.0	12.3	12.8	11.3	17.7	29.5	45.5	25.8	44.6	5.0	60	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	86.1	66.6	82.5	65.2	79.2	64.1	68.1	81.8	66.6	79.5	65.2	77.1	6.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	92.8	72.9	62.2	89.3	71.7	60.7	84.8	70.0	33.4	82.4	32.1	80.1	31.0	76.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.2	15.3	DB	4.9	91.3	6.9	3.1	-0.1	93.6	-4.2	95.4	-8.1	97.1	-13.1	99.4	(4)
(5)			WB	4.8	70.8	6.8	1.6	-0.1	72.0	-4.1	72.9	-7.9	73.8	-12.8	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	31.5	33.7	40.5	47.9	55.8	62.3	65.4	64.9	56.8	49.2	39.3	33.7	(6)	
	DBStd	13.92	8.21	8.55	7.44	6.96	6.40	6.67	5.77	5.60	5.98	6.92	6.87	7.34	(7)	
	HDD50	2418	575	458	301	119	20	1	0	0	11	101	326	506	(8)	
	HDD65	6234	1040	877	759	512	292	130	66	73	252	491	771	971	(9)	
	CDD50	1869	0	0	7	57	199	369	479	462	216	75	5	0	(10)	
	CDD65	211	0	0	0	0	7	48	80	69	7	0	0	0	(11)	
	CDH74	2252	0	0	0	6	121	533	802	701	86	3	0	0	(12)	
	CDH80	614	0	0	0	0	14	142	243	208	7	0	0	0	(13)	
	WSAvg	6.4	7.4	7.4	7.8	6.5	6.1	5.6	5.8	5.4	5.6	5.8	6.4	7.3	(14)	
	PrecAvg	35.0	1.9	1.7	2.0	2.5	4.0	4.6	4.2	3.9	2.9	2.5	2.4	2.3	(15)	
(16) Precipitation	PrecMax	45.3	4.1	3.5	6.1	5.0	6.8	7.9	7.6	7.3	5.4	5.0	5.0	5.2	(16)	
	PrecMin	26.9	0.2	0.7	0.6	0.6	0.9	2.4	1.1	1.8	1.2	0.4	0.1	0.2	(17)	
	PrecStd	4.2	1.0	0.8	1.1	1.1	1.7	1.4	1.5	1.5	1.2	1.2	1.2	1.1	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	57.6	65.8	74.8	82.0	89.3	91.2	91.1	80.8	71.7	60.4	52.2	(19)	
		MCWB	46.7	47.0	51.1	56.4	62.8	67.1	67.7	66.9	64.0	59.6	53.5	46.8	(20)	
	2%	DB	49.2	52.3	60.7	69.9	77.4	84.3	86.0	85.3	76.7	66.6	55.4	48.9	(21)	
		MCWB	44.6	44.6	48.8	53.9	61.5	66.3	66.8	66.0	62.9	57.9	49.7	45.1	(22)	
	5%	DB	45.5	48.6	56.1	65.9	73.3	80.3	82.2	81.4	72.0	62.7	51.7	46.5	(23)	
		MCWB	41.8	42.7	47.1	52.6	59.1	64.8	65.6	64.7	61.0	56.0	47.8	43.3	(24)	
	10%	DB	42.5	44.9	52.2	62.0	69.4	76.5	78.5	77.3	68.1	59.0	48.7	43.6	(25)	
		MCWB	39.8	40.5	45.3	50.8	57.2	63.2	64.1	63.7	59.1	54.3	45.5	41.1	(26)	
	0.4%	WB	48.3	48.2	52.7	58.0	65.2	70.0	70.5	69.1	66.1	61.0	53.4	48.3	(27)	
		MCDB	51.6	56.4	62.2	71.0	77.8	84.2	85.0	83.5	76.9	68.7	58.2	51.2	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.7	45.3	49.9	55.4	62.6	67.6	68.2	67.3	63.8	58.6	50.5	45.6	(29)	
		MCDB	48.7	51.1	58.4	67.7	75.0	80.5	82.1	81.2	73.7	65.0	54.5	48.6	(30)	
	5%	WB	42.0	43.0	47.9	53.2	60.3	65.6	66.6	65.9	62.0	56.7	48.2	43.4	(31)	
		MCDB	44.9	48.1	55.0	63.7	71.1	78.0	79.6	78.5	70.8	61.9	51.4	46.1	(32)	
	10%	WB	39.5	40.7	45.9	51.2	58.2	63.7	64.9	64.5	59.9	54.6	45.9	41.0	(33)	
		MCDB	41.9	44.8	51.9	60.7	67.4	74.7	76.4	75.8	66.8	58.7	48.4	43.3	(34)	
	5% DB	MDBR	9.4	12.8	16.0	19.0	19.1	19.5	19.5	19.8	17.6	14.5	10.0	8.6	(35)	
		MCDBR	13.1	19.2	24.1	27.4	27.3	27.1	27.1	27.3	24.9	20.3	15.8	12.1	(36)	
		MCWBR	9.6	12.6	14.3	14.4	13.6	12.4	11.2	11.6	13.1	12.4	11.2	9.2	(37)	
		MCWBR	12.3	17.6	21.6	24.7	24.6	25.0	24.3	24.5	22.1	18.8	14.6	11.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.291	0.315	0.348	0.381	0.392	0.398	0.399	0.394	0.369	0.341	0.313	0.287	(40)	
		taud	2.450	2.388	2.353	2.287	2.291	2.307	2.295	2.329	2.376	2.448	2.480	2.457	(41)	
	Ebn at Noon		251	267	275	275	276	274	272	268	264	254	239	237	(42)	
		Edh at Noon	20	27	33	39	40	40	40	37	32	25	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		330	591	961	1356	1580	1795	1709	1496	1099	686	369	266	(44)	
	RadStd		42	87	129	188	173	154	153	149	123	76	47	41	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	N/A	
(47)	Regional (90 neighbors)	N/A	N/A	N/A	N/A	+0.72	+0.65	N/A	N/A	+95	+31	

Nomenclature: See separate page